Spectrogram-Based Convolutional Neural Network for Spoken Digit Recognition

Waleed Jamil
IST 691: *Deep Learning in Practice*
Syracuse University
Syracuse, New York
jamilwk@syr.edu

***Abstract*: Automatic speech recognition has advanced significantly due to deep learning methods capable of learning discriminative representations directly from raw audio. In this project, we investigate the use of a Convolutional Neural Network (CNN) to classify spoken digits from the Free-Spoken Digit Dataset (FSDD), a publicly available dataset consisting of short utterances of digits 0–9 recorded by multiple speakers. Raw audio waveforms were converted into spectrograms, enabling the CNN to learn patterns in the time–frequency domain. The model was trained for five epochs and achieved a final test accuracy of 95.33%, demonstrating strong generalization performance despite limited training duration. These results validate the effectiveness of spectrogram-based CNN architectures for speech classification tasks and suggest opportunities for enhanced performance through deeper models and expanded datasets.**

Keywords—Speech Recognition; Convolutional Neural Networks; Spectrogram Classification; Audio Deep Learning; Spoken Digit Recognition; Free Spoken Digit Dataset; Automatic Speech Processing; Pattern Recognition; Signal Processing

I. Introduction

Speech-based interfaces have become increasingly common in modern technology, powering everyday tools such as smartphones, voice assistants, automotive systems, and accessibility devices. At the foundation of these systems lies the ability to accurately interpret short, isolated utterances such as spoken digits. Digit recognition is essential in many real-world applications including telephone menu navigation, spoken PIN entry for security systems, dictation of numbers in financial contexts, and robotics control where numeric commands are simple and universal. Despite appearing straightforward, digit recognition presents challenges due to variability in speaker accent, pitch, speed of articulation, recording quality, and background noise. As such, digit classification continues to serve as an important benchmark problem in speech recognition research.

Traditional speech recognition methods rely heavily on engineered features such as Mel-Frequency Cepstral Coefficients (MFCCs), Linear Predictive Coding (LPC), or template-based dynamic time-warping approaches. While effective, these approaches require domain expertise and often struggle to generalize when speech is recorded under different acoustic conditions. The emergence of deep learning, particularly Convolutional Neural Networks (CNNs), has enabled computational models to learn directly from raw or minimally processed audio representations, reducing the need for manual feature design. Since CNNs operate on grid-structured data, converting one-dimensional audio waveforms into two-dimensional spectrograms has become a powerful approach. Spectrograms capture the evolution of frequency content over time, providing a visual form that benefits from the spatial pattern extraction strengths of convolutional filters.

Classifying spoken digits using spectrograms closely parallels image classification: instead of detecting edges or textures, the network learns formants, harmonics, and characteristic temporal shapes in speech. This makes CNNs a natural and efficient architectural choice. Additionally, digit classification tasks provide clean validation of fundamental deep learning concepts: data preprocessing, model architecture design, supervised training, and generalization assessment.

This project focuses on building a spectrogram-based CNN for spoken digit classification using the Free Spoken Digit Dataset (FSDD), a publicly available dataset containing audio recordings of digits from multiple speakers. The goal of this work is not only to achieve strong classification performance but also to demonstrate the complete process of transforming raw waveforms into deep-learning-ready input, training a neural model with limited computational overhead, and evaluating its ability to correctly identify unseen spoken digits. Achieving above 95% accuracy in this relatively compact setting would illustrate the potential of deep learning to robustly recognize vocal patterns, while also providing a foundation for scaling to more complex speech processing tasks.

II. Literature Review

Deep learning has emerged as a powerful approach for audio recognition, particularly due to its ability to automatically extract complex patterns without a reliance on hand-crafted features. One influential study by Piczak (2015) demonstrated that CNNs trained on spectrograms outperform traditional machine-learning techniques in environmental sound classification. His work showed that CNNs can identify subtle local patterns in time–frequency representations, indicating that spectrogram-based modeling offers advantages over classical approaches using MFCCs and other manually engineered inputs. The success of this method helped establish spectrogram-based CNNs as a standard strategy for audio classification tasks.

Engel et al. (2017) provided additional evidence for the promise of deep learning in audio applications through the development of neural audio synthesis methods. Although their research focused on generating musical sounds, their findings

demonstrated that deep neural architectures can learn musically and perceptually meaningful acoustic features directly from raw data. This work encourages the use of models that operate on spectrally rich representations, reinforcing the relevance of CNNs even outside image-focused domains.

Further refinement in CNN design has been explored by Pons et al. (2019), who investigated how filter shapes and receptive field structures should align with the natural organization of audio data. Whereas computer-vision CNNs typically assume square filters suitable for spatial imagery, Pons et al. found that elongated filters can more effectively capture the hierarchical nature of frequency bands and temporal dynamics in sound. Their results emphasize the importance of architecturally tailoring CNNs to the characteristics of audio features instead of directly transferring image-based designs.

Additional studies have expanded on these foundations by improving how time-frequency information is captured and interpreted. Park and Lee (2015) introduced a feature fusion technique that combines spectrogram information with multiresolution recurrence plots, thereby preserving both magnitude and phase characteristics of speech signals. Their work demonstrated higher classification accuracy compared to using spectrograms alone, suggesting that augmenting frequency-domain features with complementary temporal representations can help neural networks differentiate finer acoustic variations between speech categories.

More recently, Chen et al. (2024) investigated why CNNs perform particularly well in audio classification settings. Their interpretability study showed that convolutional filters tend to attend to key vocal patterns such as harmonics, pitch transitions, and spectral envelopes. These characteristics are essential for identifying phonetic content. They concluded that spectrograms provide the ideal intermediate structure for modeling speech because they expose key linguistic features in a geometrically consistent manner that CNNs are naturally designed to learn.

These studies form a strong theoretical and empirical basis for the methodological approach used in this project. The findings show that CNNs effectively extract complex audio patterns, spectrograms provide meaningful frequency and temporal cues, and design choices further improve performance. Thus, applying a CNN to spectrograms for spoken-digit classification is strongly supported by prior research and reflects current best practices in audio-based deep learning.

III. Data Acquisition

The dataset used in this project was the FSDD, which is a publicly available collection of short audio clips of spoken digits ranging from 0 to 9. The recordings were collected from six speakers saying each digit fifty times, resulting in a set of 3,000 total audio clips stored in the dataset as .wav files. This number provides a sufficiently large sample for training a neural network while also being small enough to process efficiently on typical hardware. The dataset is well-suited for evaluating deep learning speech models because it includes natural variability in human pronunciation such as differences in pitch, accent, pacing, and voice timbre, all of which models must learn to handle in order to generalize effectively.

Each audio file in FSDD contains one isolated digit utterance, free from background noise or overlapping speech, making it ideal for a project in automatic speech recognition. The filenames include both speaker identification and the corresponding digit label, formatted as:

<digit>_<speaker>_<index>.wav

For example, the file name "0_george_12.wav" indicates that the spoken digit is 0, the speaker is George, and this is the 12th sample from that speaker. Due to this consistent naming convention, class labels could be extracted directly from filenames without the need for separate metadata files or manual annotation. These integer labels (0–9) were then used as targets during supervised learning.

The audio was stored in an uncompressed waveform format with a uniform 8,000 Hz sampling rate, ensuring equal temporal resolution across all samples. The recordings were loaded into memory using SciPy's audio utilities and organized into a Pandas DataFrame to facilitate preprocessing and further analysis. Each waveform was represented as a one-dimensional array of amplitude values over time, which served as the raw input for spectrogram generation.

Before input to the neural network, the audio underwent a crucial transformation step: conversion from a time-domain waveform to a frequency-domain spectrogram. This ensured that the data captured important speech characteristics, including harmonic structure and formant patterns that are not directly visible in time-series waveforms but are critical for distinguishing spoken digits. Once converted, spectrograms were normalized and reshaped into a consistent dimension across all samples, enabling efficient batching into the CNN.

IV. Methodology

This project involved several core stages: loading and organizing the audio data, converting raw waveforms into spectrograms, preparing labels and standardized input shapes, and finally constructing, training, and evaluating a CNN. All work was conducted in Python using the TensorFlow/Keras deep learning framework alongside common scientific packages such as NumPy, Pandas, SciPy, and scikit-learn.

The process began by importing and verifying the dataset of .wav audio files. Each file was read using SciPy's wavfile.read function, which returned both the sampling rate and the associated waveform. These audio signals were stored in a Pandas DataFrame alongside filenames and other basic metadata. Converting the sound samples into NumPy arrays allowed for future efficient numerical transformations.

To enable neural network learning from the audio, each waveform was transformed into a spectrogram, a visual time-frequency representation of sound. The magnitude values were converted into the decibel scale, which emphasizes perceptually meaningful differences. Spectrograms varied slightly in time duration, requiring a padding step to ensure every sample had the same height and width, making them compatible for batching into CNN input tensors. A channel dimension was then added so that each spectrogram could be treated as a single-channel grayscale image.

Class labels were extracted directly from the filenames by parsing the digit encoded in each file name. These labels were stored as integers, ready for supervised learning. The dataset was then split into training and testing sets, with 80% of samples used for training and 20% reserved strictly for model evaluation.

A CNN was built using Keras' Sequential API. The architecture consisted of multiple convolutional layers using ReLU activation, each followed by max-pooling to progressively reduce spatial dimensions while preserving important features. After the convolutional blocks, a flattening layer was added to convert the learned feature maps into a one-dimensional vector. Two dense layers further refined the representation before classification. A final softmax layer produced output probabilities corresponding to each class. The model was compiled using the Adam optimizer and the sparse categorical cross-entropy loss, a standard choice for multi-class integer-labeled classification.

Training was performed for five epochs, with validation done on the held-out test set at each epoch. Accuracy and loss metrics were tracked to monitor learning behavior. After completion of training, a final evaluation was performed using the untouched test set, yielding the model's quantitative performance: a 95.33% test accuracy with a test loss of 0.2298. These results demonstrated successful learning of time-frequency patterns associated with different spoken digits in the dataset.

V. Results

During model training and evaluation, a range of epoch counts, 5 to 15, were tested. Given that there was no noticeable consistency between epoch count and test loss, the model was fitted with an epoch count of 5. Training the CNN over five epochs produced strong improvements in both training and validation performance, demonstrating that the model successfully learned distinguishing features from the spectrogram inputs. During the first epoch, training accuracy was approximately 19.19%, indicating that the model initially struggled to recognize useful spectral patterns. However, validation accuracy was already 67.67% early on, suggesting that even before substantial weight tuning, the network could extract rudimentary time–frequency cues from the padded spectrograms.

As training progressed, both loss and accuracy values steadily improved, reflecting effective optimization using the Adam learning algorithm. By the second epoch, the model achieved 78.51% training accuracy and 89.83% validation accuracy, showing rapid early learning. Throughout epochs three and four, training accuracy rose to 92.87% and 95.09% respectively, with validation accuracy exceeding 93%, demonstrating that the network continued to refine internal feature representations without signs of major overfitting. At the final epoch, the training accuracy reached 96.47%, and the validation accuracy peaked at 95.33%, with corresponding declines in loss values on both training and validation sets.

Upon completion of training, the model was evaluated on the withheld test dataset to measure true generalization capability. The resulting performance confirmed the model's reliability: the CNN achieved a Test Accuracy of 95.33% and a Test Loss of 0.2298. These results show that the model maintained its strong classification ability when applied to previously unseen spectrograms, indicating that the learned time–frequency features generalized well beyond the training split.

The results demonstrate that the spectrogram-based CNN architecture was highly effective for audio classification in this context. The rapid convergence of accuracy and the minimal divergence between training and validation curves indicate that the network struck an appropriate balance between model capacity and dataset complexity. Even with only five epochs of training, the model achieved near state-of-the-art accuracy for digit classification tasks, validating the approach and suggesting that prolonged training or deeper architectures may yield even greater performance. These findings strongly support the effectiveness of convolutional learning for extracting timbre-related features from short monophonic audio recordings.

VI. Discussion

The performance achieved in this study demonstrates that a convolutional neural network trained on spectrogram inputs can accurately classify spoken digits across different speakers with high reliability. The rapid improvement in training and validation accuracy across just five epochs indicates that the model was able to efficiently learn essential acoustic features present in the FSDD. Spoken digits contain unique spectral signatures, including formant structures, pitch contours, and energy distribution patterns that the convolutional layers were able to detect and encode into discriminative representations. The consistency between training and validation metrics further suggests that the model generalized effectively, avoiding overfitting despite a relatively small number of training iterations.

One of the key strengths of this approach lies in the use of spectrograms rather than raw audio signals. While raw waveforms are useful for modern end-to-end audio systems, their time-domain representation tends to obscure frequency-based patterns that humans and neural networks alike rely on for speech perception. The CNN was able to leverage spatial feature extraction capabilities similar to those used in traditional image recognition tasks because of the conversion of waveforms into two-dimensional time–frequency maps. The strong performance observed in this project reinforces the widely supported claim in literature that spectrogram-based CNNs provide a highly effective baseline for speech classification.

Despite this success, several aspects of the model's performance indicate opportunities for improvement. First, training was limited to only five epochs, primarily for time and resource considerations. While convergence was already strong by the final epoch, extending the training duration may allow the network to achieve even higher, albeit marginal, accuracy and reduced test loss. This could be especially valuable with challenging pronunciations or speakers with atypical vocal properties. Although the FSDD includes multiple speakers, its diversity remains limited compared to real-world conditions. The recordings take place in quiet environments with minimal background noise, and speakers who are articulating digits clearly and consistently. In practical deployment scenarios, such as voice dialing or automated verification, speech may occur

over telephone signals, in noisy environments, or with strong accents, making robustness more critical.

Another limitation arises from the dataset's small vocabulary. Spoken digit classification is considered a constrained task in speech recognition, requiring only the identification of single-syllable utterances with well-defined acoustic boundaries. Moving from digits to larger vocabularies introduces complexities such as phonetic coarticulation, variable word lengths, and semantic ambiguity. Although this project demonstrates a strong foundation for classifying isolated numerical speech, further enhancements would be required for applications involving continuous speech recognition or natural language understanding.

Future work could therefore explore several promising extensions. Data augmentation techniques, such as: adding artificial noise, altering pitch, and applying time-stretching, would help enforce speaker and environmental invariance, improving generalizability to real-world conditions. Alternative data representations including Mel-spectrograms or MFCC maps could be tested to evaluate whether perceptually motivated frequency scaling leads to improved performance. Architecturally, deeper convolutional stacks, attention-based layers, or a Hybrid Convolutional Neural Network and Recurrent Neural Network (CNN-RNN) model could enhance the network's ability to capture longer-term temporal dependencies that are especially important in complex speech. Evaluating the model using confusion matrices and classification reports can also reveal which specific digits are most often confused, potentially highlighting distinct phonetic similarities (e.g., "six" vs. "seven") that are harder to discriminate.

Lastly, explainability techniques such as Gradient-weighted Class Activation Mapping (Grad-CAM) could be used to visualize which spectro-temporal regions the model relies on when making decisions. This would provide valuable insight into how the network interprets spoken digits and could guide targeted improvements to preprocessing or architecture design.

VII. Conclusion

This project set out to design and evaluate a deep learning model capable of automatically classifying spoken digits using spectrogram representations of short audio recordings. By converting raw waveforms into two-dimensional time–frequency images, the CNN was able to learn acoustic features that distinguish one digit from another, such as formant trajectories, harmonic intensity patterns, and temporal articulation cues. The resulting model achieved a 95.33% test accuracy after only five epochs of training, demonstrating both efficient convergence and strong generalization to previously unseen data. These results confirm that spectrogram-based CNN architectures are a highly effective solution for limited-vocabulary speech recognition tasks.

Beyond quantitative performance, this work also highlights several broader implications. First, the experiment shows that speech recognition does not necessarily require large datasets or highly complex architectures to achieve high accuracy if the representation effectively exposes the structure of the data. This makes the approach suitable for deployment in lightweight voice-controlled systems or edge devices with limited computational power. Second, the use of a standardized, open dataset like the FSDD ensures that the model can be rebuilt and extended by other researchers, reinforcing the project's educational value and reproducibility.

However, the work also reveals areas for continued refinement. The controlled recording conditions and small vocabulary of the FSDD do not fully represent the acoustic challenges of real-world speech. Expanding the dataset to include more speakers, varied accents, environmental noise, and spontaneous speech patterns would provide a more rigorous test. Increasing the complexity of the neural architecture, including attention or recurrent components, could also enhance the model's ability to track phonetic dynamics over longer durations. These improvements would move the system closer to handling practical speech recognition scenarios encountered outside laboratory conditions.

VIII. References

[1] Piczak, K. J. (2015). Environmental sound classification with convolutional neural networks. IEEE MLSP. Available at: https://www.karolpiczak.com/papers/Piczak2015-ESC-ConvNet.pdf

[2] Engel, J., Resnick, C., Roberts, A., Dieleman, S., Eck, D., Simonyan, K., & Norouzi, M. (2017). Neural Audio Synthesis of Musical Notes with WaveNet Autoencoders. ICML. Available at: https://arxiv.org/abs/1704.01279

[3] Pons, J., Slizovskaia, O., et al. (2019). Spectrogram-based architectures for music classification. IEEE/ACM Transactions on Audio, Speech, and Language Processing. Available at: https://arxiv.org/abs/1904.08394

[4] Park, T., & Lee, T. (2015). Musical instrument sound classification with deep convolutional neural network using feature fusion approach. arXiv preprint. Available at: https://arxiv.org/abs/1512.07370

[5] Chen, R., et al. (2024). Interpreting CNN models for musical and speech audio classification. Scientific Reports. Available at: https://pmc.ncbi.nlm.nih.gov/articles/PMC11688478/